Corrected and conduct of chapter 2

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $x^2 - 6x + 9$. *Action:* Reconstitute the product of two pairs.
2. $(x - 4)(x + 4)$. *Action:* Use the difference of two squares out of two examples.
3. x = 2 or $x = -5$. *Action:* Verbate the null product rule and its hypothesis.
4. $x = -3$ or $x = 3$. *Action:* Distinguish the positive square root from the equation solutions.
5. Negative before 3, zero in 3, positive after 3. *Action:* Draw the right and then make the signboard.
6. 4 reached only in 1. *Action:* From the positive character of a square.

Research activity

Positive lengths x and y check $x + y$ = 10 and xy = 24. They are the roots of $t^2 - 10t + 24 = (t - 4)(t - 6)$. Dimensions 4 m and 6 m; control of the area and perimeter.

Immediate applications

1. $g(x) = (x + 1)^2 - 4 = (x - 1)(x + 3)$. Roots 1 and -3, minimum -4 in -1.
2. The first is 3($x - 1)^2$ = 0, therefore 1. For the second one, $-16 < 0$: no real solution.
3. $f(x) = a(x - 1)(x - 4)$; $4a = -8$, so $f(x) = -2(x - 1)(x - 4)$.
4. $(x - 5)(x + 1) < 0$ gives] -1; 5[. The summit of $x(8 - x)$ is in 4, out of the domain; on [0; 3], the function grows and its maximum is 15 in 3.

Exercises: approach and results

Exercise 2.1 · Complete a square. $x^2 - 8x + 7 = (x - 4)^2 - 9 = (x - 1)(x -$ (7). The coefficient dominant is positive; the trinomial is negative or zero between the roots: [1; 7].

Exercise 2.2 · Discuss without heavy calculation. $x^2 + 4 > 0$ always, without root. ($x + 2)^2 \geq 0$, double root -2. For the last, roots -4 and 1 ; positive on] $-$ 4; 1[, negative on the outside, zero on the Boundaries.

Exercise 2.3 · Choose a strategy. Identity: $x = \pm 3$. Sum and product: 4 and 5.
third, no obvious whole root: $x = \frac{3 \pm \sqrt{17}}{4}$. The discriminant is suitable in
all cases, but is not always the most effective.

Exercise 2.4 · Find a parabola. $f(x) = a(x + 2)(x - 3$. In 1: $-6a$ = 12, therefore $a = -2$; $f(x) = -2x^2 + 2x$ + 12, where $f(0) = 12$.

Exercise 2.5 · A vegetable garden against a wall. The two widths use 2 x meters; length $20 - 2x$. Domain [0; 10] for the function,]0; 10[for an undegenerated rectangle. $A(x) = 20x - 2x^2 = 50 - 2(x-5)^2$. Maximum 50 m ^{2}dimensions 5 m and 10 m.

Exercise 2.6 · A parameter. $\Delta = 4 - 4m$. Two roots if $m < 1$; double root 1 if $m = 1$; none if $m > 1$. The parameter m remains fixed when resolving in x.

Exercise 2.7 · Link to the sequences. $u_n = (n-1)(n-4)$; negative for 1 $< n <$ 4. Like n is integer, the only rows are 2 and 3. All ranks are not the integer real range.

Exercise 2.8 · Deferred review. The minimum is -5 in -3. The equation gives ($x+3)^2 \leq 1$, let $-1 \leq x+3 \leq 1$, therefore $[-4; -2]$.

CQFD, exercise 43 p. 99, questions a and c. a: obvious root 1, product -30, so second root -30. c: write 4 $x^2 + 9x$ + 5 ; root -1, product $\frac{5}{4}$, so second root $-\frac{5}{4}$. Check the amounts, respectively -29 and $-\frac{9}{4}$.

CQFD, exercise 49 p. 99, questions a and b. Numbers are the roots of $t^2 - st + p$. a : $t^2 - t$ + 1, discriminant -3 : no actual torque. b : $t^2 + 60t$ + 756, discriminant 576 = 24 ^{2} : -18 and -42. Their sum is equal to -60 and their products 756.

MCQ: answers and reasons

MCQ 1.

1. C. Cancel each factor.

2. A. $4 - 12 = -8$.

3. D. The square cancels in 2, value 5.

4. B. $-\frac{b}{a} = \frac{6}{2} = 3$.

5. C. Positive dominant coefficient, strict inequality.

MCQ 2.

1. Remarkable identity.

2. A. $\frac{c}{a} = -\frac{2}{3}$.

3. C. The function grows up to 2; on this domain, right edge 1.

4. B. For non-zero, the null discriminant characterizes the double root.

5. D. The multiplier coefficient is free and not zero.

True or false

1. False: x^2 + 1 is strictly positive.

2. False without hypothesis on a; $-(x-1)(x-2)$ is positive between 1 and 2.

3. True: the square of a real is positive or null; equality only in 2.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.